# Experiment - 10

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**(A)**
Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction.
If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints.
The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

**Solution:**

```
CREATE OR REPLACE PROCEDURE insert_students()
LANGUAGE PLPGSQL
AS $$
BEGIN

    -- Insert 1
    BEGIN
        INSERT INTO students(name, age, class)
        VALUES ('Anisha', 16, 8);

        RAISE NOTICE 'Anisha inserted successfully';

    EXCEPTION WHEN OTHERS THEN
        RAISE NOTICE 'Anisha insertion failed. Rolled back.';
    END;


    -- Insert 2
    BEGIN
```

```
    INSERT INTO students(name, age, class)
    VALUES ('Neha', 17, 8);

    RAISE NOTICE 'Neha inserted successfully';

  EXCEPTION WHEN OTHERS THEN
    RAISE NOTICE 'Neha insertion failed. Rolled back.';
  END;


  -- Insert 3 - INVALID
  BEGIN
    INSERT INTO students(name, age, class)
    VALUES ('Anisha', 19, 9);

    RAISE NOTICE 'Anisha inserted successfully';

  EXCEPTION WHEN OTHERS THEN
    RAISE NOTICE 'Anisha insertion failed. Only this insertion was rolled back.';
  END;


  -- Insert 4
  BEGIN
    INSERT INTO students(name, age, class)
    VALUES ('Mayank', 19, 9);

    RAISE NOTICE 'Mayank inserted successfully';

  EXCEPTION WHEN OTHERS THEN
    RAISE NOTICE 'Mayank insertion failed. Rolled back.';
  END;


  RAISE NOTICE 'Transaction Successfully Done';

END;
$$;
```

```
BEGIN;

CALL insert_students();

COMMIT;

SELECT * FROM students;
```